

Assignment: Day 3

Introduction to IP Addressing

Q1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Ans:

IPv4	IPv6
Address length: 32 bits	Address length: 128 bits
Format: four decimal numbers separated by dots, e.g. 192.168.1.10	Format: eight groups of hexadecimal digits separated by colons, e.g. 2001:0db8:0000:0000:0000:0000:0000:0001 (short form 2001:db8::1)
Address space: about 4.3 billion addresses, which is not enough for today's devices	Address space: about 3.4×10^{38} addresses, enough for every device
Header: 20 to 60 bytes, variable size with many fields	Header: fixed 40 bytes, simpler and faster for routers to process
NAT: widely needed because public addresses are scarce	NAT: not needed because every device can have its own public address
Configuration: manual or DHCP	Configuration: automatic (SLAAC) as well as DHCPv6
Communication types: unicast, multicast and broadcast	Communication types: unicast, multicast and anycast (no broadcast)
Security: IPsec is optional	Security: IPsec support is built into the protocol

Q2. You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Ans: The network is 192.168.10.0/24, so the subnet mask is 255.255.255.0. I kept 192.168.10.1 for the WiFi router (the default gateway) and gave each device a different address from the remaining usable range (192.168.10.2 to 192.168.10.254).

Device	Static IP address	Subnet mask	Default gateway
Laptop	192 . 168 . 10 . 10	255 . 255 . 255 . 0	192 . 168 . 10 . 1
Phone	192 . 168 . 10 . 11	255 . 255 . 255 . 0	192 . 168 . 10 . 1
Smart TV	192 . 168 . 10 . 12	255 . 255 . 255 . 0	192 . 168 . 10 . 1
Printer	192 . 168 . 10 . 13	255 . 255 . 255 . 0	192 . 168 . 10 . 1

Q3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Ans: 192.168.1.0 is a Class C network with a default mask of /24. Using /27 means 3 extra bits are borrowed for subnetting and 5 bits are left for hosts.

/27 in dotted form	255 . 255 . 255 . 224
Number of subnets	$2^3 = 8$ subnets
Hosts per subnet	$2^5 - 2 = 30$ usable hosts (2 addresses are kept for network and broadcast)
Block size	$256 - 224 = 32$, so subnets start at 0, 32, 64, 96, 128, 160, 192, 224

First three subnets:

Subnet	Network address	Usable host range	Broadcast address
1	192.168.1.0/27	192.168.1.1 to 192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33 to 192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65 to 192.168.1.94	192.168.1.95

Q4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Ans: Step 1: Convert the mask to CIDR. 255.255.255.0 has 24 ones (8 + 8 + 8), so the mask is /24.

Step 2: Network address = IP address AND subnet mask. The first three octets have a mask of 255 (all ones), so they stay the same. Only the last octet changes:

	Octet 1	Octet 2	Octet 3	Octet 4
IP address 10.0.5.17	00001010	00000000	00000101	00010001
Subnet mask 255.255.255.0	11111111	11111111	11111111	00000000
AND result	00001010	00000000	00000101	00000000
In decimal	10	0	5	0

Step 3: Broadcast address = network address with all 8 host bits set to 1, so the last octet becomes 255.

Result	Value
Network address	10.0.5.0
Broadcast address	10.0.5.255
Valid host range	10.0.5.1 to 10.0.5.254
Usable hosts	$2^8 - 2 = 254$

Q5. Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Ans: Input given to the calculator: IP address 10.0.5.17, subnet mask 255.255.255.0 (/24).
Screenshot of the result:

Subnet Calculator

IPv4 result

Input

IP address

10.0.5.17

Subnet mask

255.255.255.0 (/24)

Result

Network address	10.0.5.0
Broadcast address	10.0.5.255
Usable host range	10.0.5.1 - 10.0.5.254
Total addresses	256
Usable hosts	254
CIDR notation	10.0.5.0/24
Wildcard mask	0.0.0.255
IP class / type	Class A (private range)
Network (binary)	00001010.00000000.00000101.00000000
Mask (binary)	11111111.11111111.11111111.00000000

Result: My manual answer matched the calculator's output for the network address, broadcast address, host range and number of hosts.